

Extraction of Dust Temperature and Column Density Maps from Herschel Data

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Abstract

This project estimates the dust temperature and molecular hydrogen column density of a far infrared and submillimetre emitting cloud using Herschel maps at 160 μm , 250 μm , 350 μm , and 500 μm . The main idea is that cold interstellar dust emits thermal radiation in the far infrared and submillimetre regime. By measuring the surface brightness at several wavelengths, the spectral energy distribution of each sky pixel can be fitted using a modified blackbody model. Before fitting, the maps must be placed on the same physical basis. Therefore, the data were converted to common surface brightness units, convolved to a common 500 μm beam, reprojected to the same pixel grid, and masked to retain only the common valid region. A pixel by pixel modified blackbody fit was then performed with a fixed emissivity index $\beta = 2$. The final valid sample contained 4896 pixels. The fitted dust temperature ranges from 6.868 K to 60.000 K, with a median value of 18.423 K. The derived molecular hydrogen column density ranges from $1.975 \times 10^{18} \text{ cm}^{-2}$ to $1.345 \times 10^{22} \text{ cm}^{-2}$, with a median value of $1.276 \times 10^{21} \text{ cm}^{-2}$. The maximum temperature reaches the imposed upper fitting bound and should therefore be interpreted cautiously.

1 Introduction

Molecular clouds are cold, dusty regions of the interstellar medium where stars can form. Since such clouds are optically dark at visible wavelengths, their internal structure is often studied through thermal emission from dust grains. Cold dust with temperatures of order 10 to 30 K emits most strongly in the far infrared and submillimetre regime. Herschel observations at 160 μm , 250 μm , 350 μm , and 500 μm are therefore well suited for tracing cold dust emission.

The goal of this project is to convert multi wavelength Herschel imaging data into two physical maps:

$$T_d(x, y), \tag{1}$$

the dust temperature map, and

$$N(\text{H}_2)(x, y), \quad (2)$$

the molecular hydrogen column density map.

Column density is the number of particles integrated along the line of sight:

$$N = \int n \, ds. \quad (3)$$

It has units of cm^{-2} . For a molecular cloud, the relevant quantity is usually $N(\text{H}_2)$, since most hydrogen is in molecular form.

The challenge is that brightness alone does not directly give column density. A pixel may appear bright because the dust is warm, because there is more material along the line of sight, or both. Multi wavelength fitting solves this problem. The shape of the spectral energy distribution constrains the dust temperature, while the amplitude constrains the amount of emitting material.

2 Physical Background

2.1 Far infrared and submillimetre emission

The Herschel wavelengths used here cover the far infrared and submillimetre regime. Roughly, far infrared wavelengths lie from tens to a few hundred microns, while the submillimetre regime extends from a few hundred microns to about one millimetre. These wavelengths lie between infrared astronomy and radio astronomy. For cold dust, this is the most useful part of the spectrum.

Using Wien’s displacement law,

$$\lambda_{\text{peak}} \approx \frac{2898 \, \mu\text{m K}}{T}, \quad (4)$$

a dust temperature of 15 K gives

$$\lambda_{\text{peak}} \approx 193 \, \mu\text{m}. \quad (5)$$

Thus the Herschel bands naturally sample the thermal emission from cold dust.

2.2 Radiative transfer

The specific intensity emitted by dust at frequency ν can be written as

$$I_\nu = B_\nu(T_d) (1 - e^{-\tau_\nu}), \quad (6)$$

where $B_\nu(T_d)$ is the Planck function and τ_ν is the optical depth. The Planck function is

$$B_\nu(T_d) = \frac{2h\nu^3}{c^2} \frac{1}{\exp(h\nu/k_B T_d) - 1}. \quad (7)$$

For far infrared and submillimetre dust emission in many molecular cloud regions, the optical depth is small:

$$\tau_\nu \ll 1. \quad (8)$$

Therefore,

$$1 - e^{-\tau_\nu} \approx \tau_\nu, \quad (9)$$

and Eq. 6 becomes

$$I_\nu \approx B_\nu(T_d)\tau_\nu. \quad (10)$$

This approximation is important because it directly connects the observed intensity to the dust column.

2.3 Column density from dust emission

The optical depth is related to the mass surface density Σ through

$$\tau_\nu = \kappa_\nu \Sigma, \quad (11)$$

where κ_ν is the dust opacity per unit total gas plus dust mass. The mass surface density can be written in terms of molecular hydrogen column density as

$$\Sigma = \mu_{H_2} m_H N(H_2), \quad (12)$$

where m_H is the hydrogen atom mass and $\mu_{H_2} = 2.8$ is the molecular weight per hydrogen molecule including helium and metals.

Therefore,

$$\tau_\nu = \kappa_\nu \mu_{H_2} m_H N(H_2). \quad (13)$$

Substituting into Eq. 10 gives

$$I_\nu = B_\nu(T_d) \kappa_\nu \mu_{H_2} m_H N(H_2). \quad (14)$$

The opacity was assumed to follow a power law:

$$\kappa_\nu = \kappa_0 \left(\frac{\nu}{\nu_0} \right)^\beta. \quad (15)$$

In this work,

$$\beta = 2, \quad \kappa_0 = 0.1 \text{ cm}^2 \text{ g}^{-1}, \quad \nu_0 = 10^{12} \text{ Hz.} \quad (16)$$

Thus the fitted parameters at every pixel are T_d and $N(\text{H}_2)$.

3 Data and Computational Workflow

The input data consist of Herschel maps at four wavelengths:

$$160, 250, 350, 500 \text{ } \mu\text{m}. \quad (17)$$

The original files contained the science maps in the FITS extension named **image**. The PACS 160 μm map was in Jy pixel^{-1} , while the SPIRE 250 μm , 350 μm , and 500 μm maps were in Jy beam^{-1} .

The full reduction and fitting workflow was divided into six scripts. This modular structure made it easier to inspect each step separately and check the intermediate products.

Table 1: Summary of scripts used in the analysis.

Script	Purpose
00_inspect_fits.py	Inspected all raw FITS files, listed HDUs, identified the science image extension, checked units, image shapes, WCS information, and kernel properties.
01_extract_convert_preview.py	Extracted the science image from each FITS file and converted the maps to common surface brightness units of MJy sr^{-1} . It also produced preview plots of the four input maps.
02_convolve_to_500beam.py	Convolved the 160 μm , 250 μm , and 350 μm maps to the 500 μm beam using the provided convolution kernels. The 500 μm map was kept unchanged as the target beam.
03_reproject_to_500grid.py	Reprojected all convolved maps onto the 500 μm WCS grid, so that the same pixel index corresponded to the same sky position in every band.
04_make_common_mask.py	Created a common valid mask by keeping only pixels that were finite and positive in all four Herschel bands. The valid region contained 4896 pixels.
05_fit_modified_blackbody.py	Performed pixel by pixel modified blackbody fitting using the four Herschel intensities. It generated the dust temperature map, column density map, logarithmic column density map, and relative residual map.

4 Image Preparation Steps

4.1 Unit conversion

SED fitting compares intensities at different wavelengths. Therefore all maps must be in the same physical unit. A value in Jy beam^{-1} depends on the telescope beam size, while a value in Jy pixel^{-1} depends on the pixel area. These are not directly comparable.

For the PACS 160 μm map, the conversion was

$$I_\nu(\text{Jy sr}^{-1}) = \frac{F_\nu(\text{Jy pixel}^{-1})}{\Omega_{\text{pixel}}}, \quad (18)$$

where

$$\Omega_{\text{pixel}} = |\Delta x \Delta y| \left(\frac{\pi}{180} \right)^2. \quad (19)$$

For the SPIRE maps, the conversion was

$$I_\nu(\text{Jy sr}^{-1}) = \frac{F_\nu(\text{Jy beam}^{-1})}{\Omega_{\text{beam}}}. \quad (20)$$

For a Gaussian beam,

$$\Omega_{\text{beam}} = 1.1331 \theta_{\text{FWHM}}^2, \quad (21)$$

where θ_{FWHM} is in radians. The final unit was then converted to

$$\text{MJy sr}^{-1}. \quad (22)$$

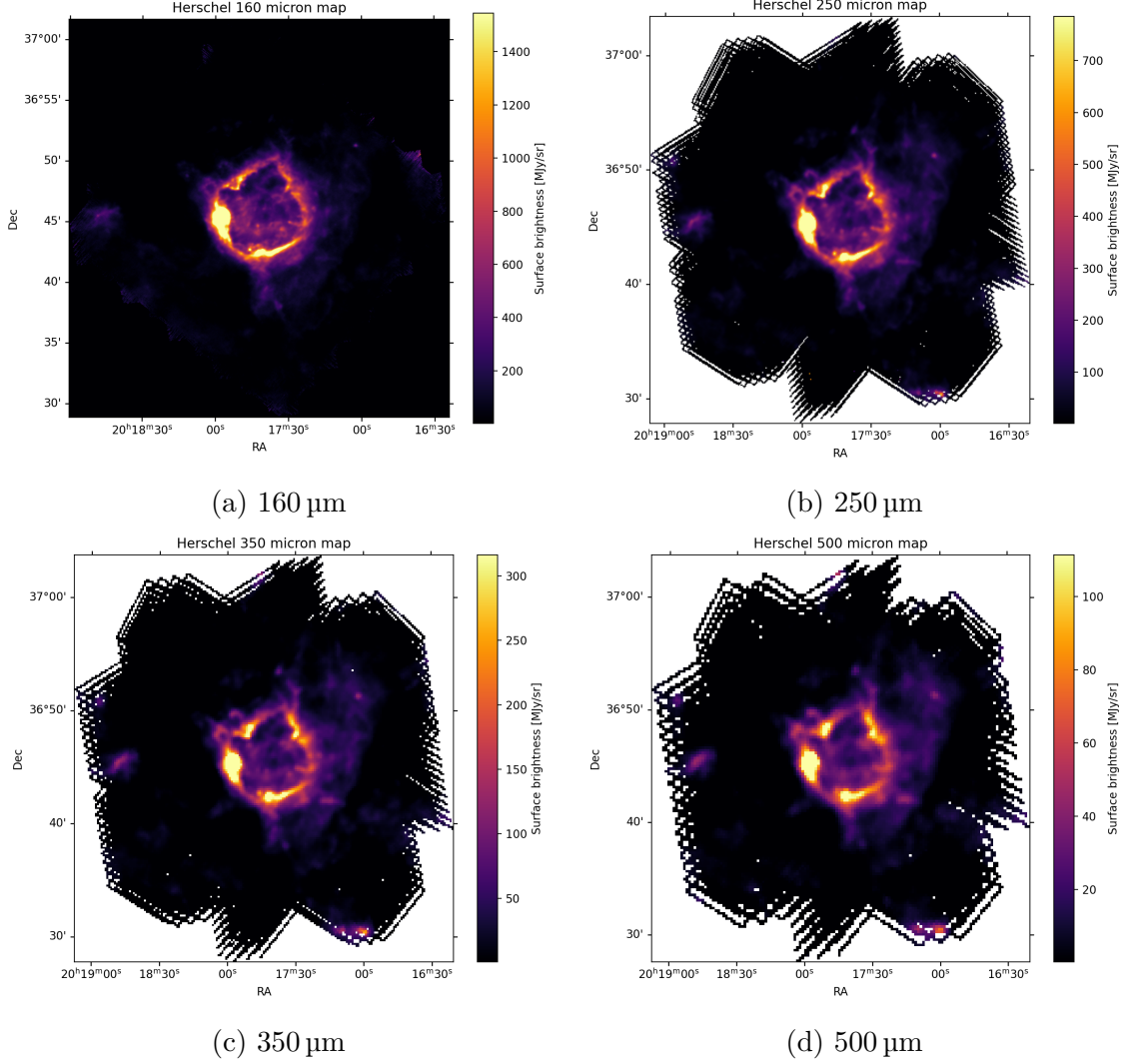


Figure 1: Raw Herschel maps after conversion to common surface brightness units of MJy sr^{-1} . The maps still have different beam sizes and pixel grids at this stage.

4.2 Beam matching

The angular resolution of a diffraction limited telescope depends on wavelength:

$$\theta \sim \frac{\lambda}{D}. \quad (23)$$

Since λ is larger at 500 μm than at 160 μm , the 500 μm map has the poorest angular resolution. Directly comparing the native maps would be physically inconsistent because each band would average the sky over a different angular scale.

To solve this, all maps were convolved to the 500 μm beam. The provided kernels were used for

$$160 \rightarrow 500, \quad 250 \rightarrow 500, \quad 350 \rightarrow 500. \quad (24)$$

The 500 μm image was kept unchanged.

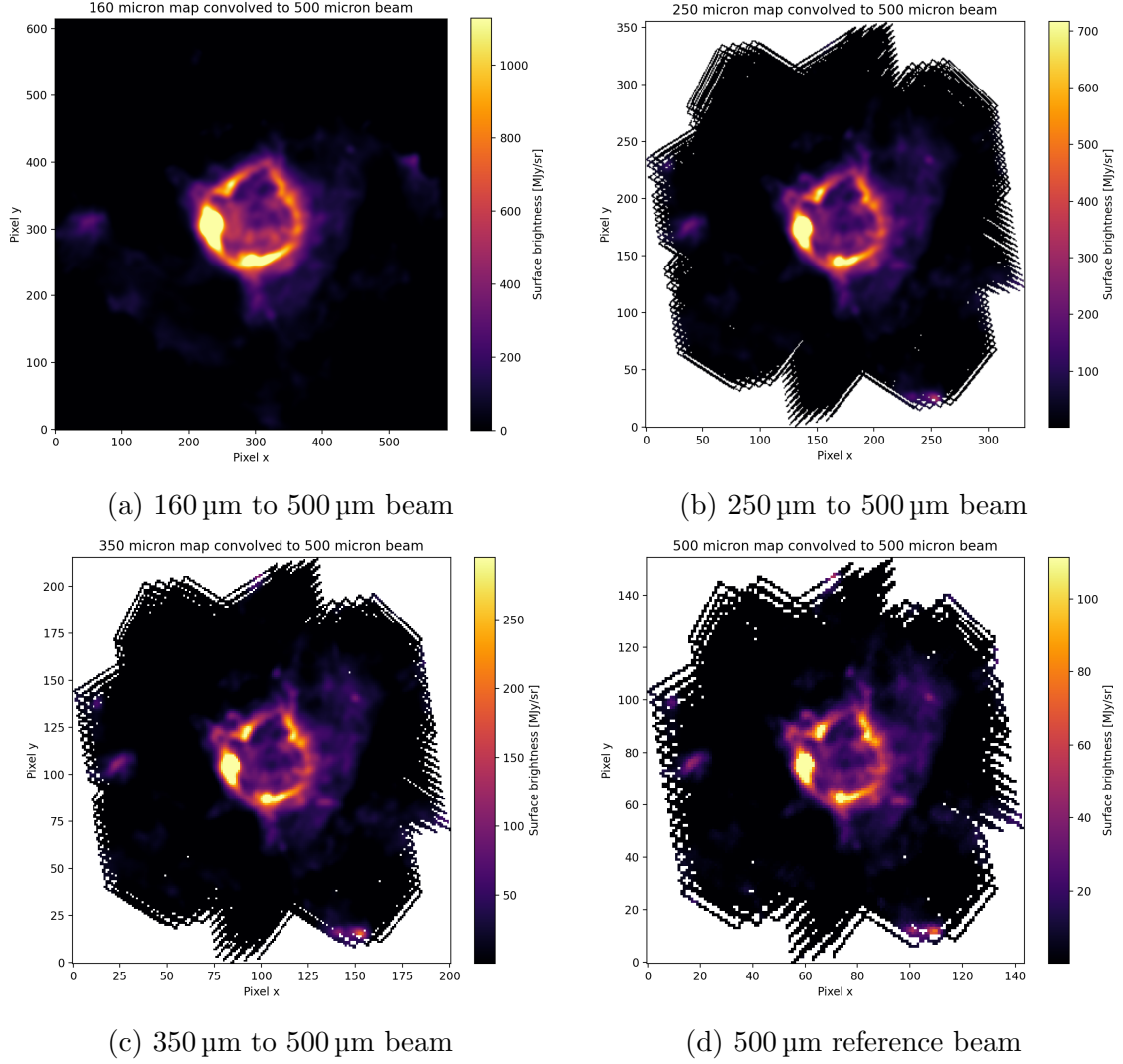


Figure 2: Maps after convolution to the common 500 μm beam. Beam matching ensures that all wavelengths probe the same effective angular scale before SED fitting.

4.3 Reprojection to a common grid

After beam matching, the maps still had different pixel grids. The 500 μm map was used as the reference grid because it has the poorest native resolution and is also the target beam. All other bands were reprojected to this grid. After reprojection, every map had the same shape:

$$155 \times 144. \quad (25)$$

This makes it meaningful to compare

$$I_{160}(i, j), I_{250}(i, j), I_{350}(i, j), I_{500}(i, j) \quad (26)$$

at the same pixel index.

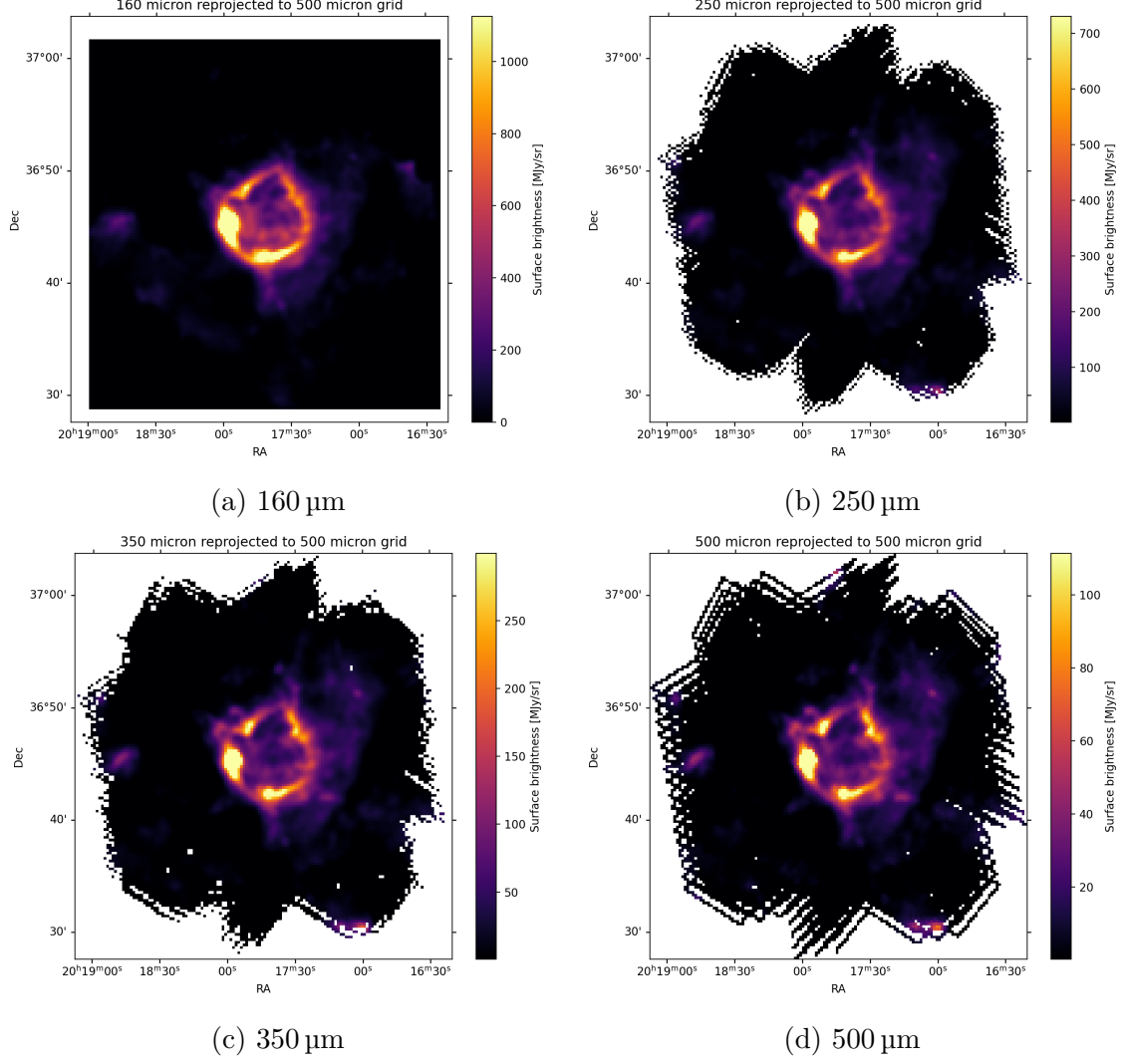


Figure 3: Maps after reprojection to the common 500 μm grid. At this stage all maps have the same beam and the same pixel grid.

4.4 Common valid mask

Some pixels were invalid or non positive in one or more bands. Since the modified blackbody fit requires four positive intensities at each pixel, a common valid mask was created:

$$I_{160} > 0, \quad I_{250} > 0, \quad I_{350} > 0, \quad I_{500} > 0. \quad (27)$$

The mask retained 4896 pixels out of 22320 total pixels, corresponding to a valid fraction of 0.2194.

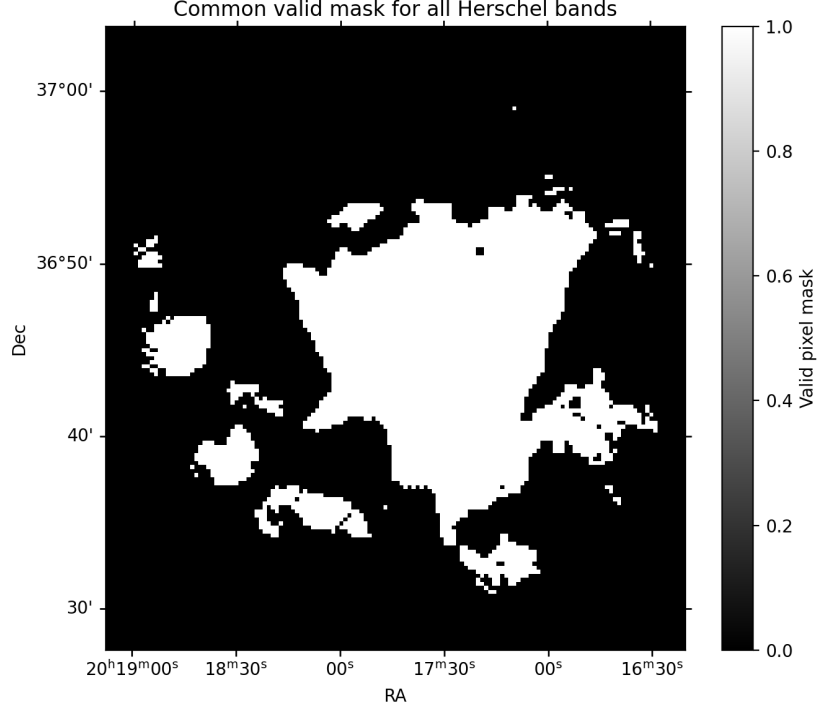


Figure 4: Common valid mask used for SED fitting. Only pixels that were finite and positive in all four bands were retained.

5 SED Fitting Method

For each valid pixel, the observed data vector was

$$[I_{160}, I_{250}, I_{350}, I_{500}]. \quad (28)$$

This was fitted using the modified blackbody model:

$$I_\nu = B_\nu(T_d) \kappa_0 \left(\frac{\nu}{\nu_0} \right)^\beta \mu_{\text{H}_2} m_{\text{H}} N(\text{H}_2). \quad (29)$$

The fitted parameters were

$$T_d \quad \text{and} \quad N(\text{H}_2). \quad (30)$$

Instead of fitting $N(\text{H}_2)$ directly, the script fitted $\log_{10} N(\text{H}_2)$, which is numerically more stable. The fit bounds were

$$5 \text{ K} \leq T_d \leq 60 \text{ K}, \quad (31)$$

and

$$18 \leq \log_{10} N(\text{H}_2) \leq 25. \quad (32)$$

The emissivity index was fixed at

$$\beta = 2. \quad (33)$$

An example SED fit is shown in Fig. 5. The observed intensities at four wavelengths are fitted by a single modified blackbody curve. The temperature is mainly constrained by the shape of the SED, while the column density is mainly controlled by the amplitude.

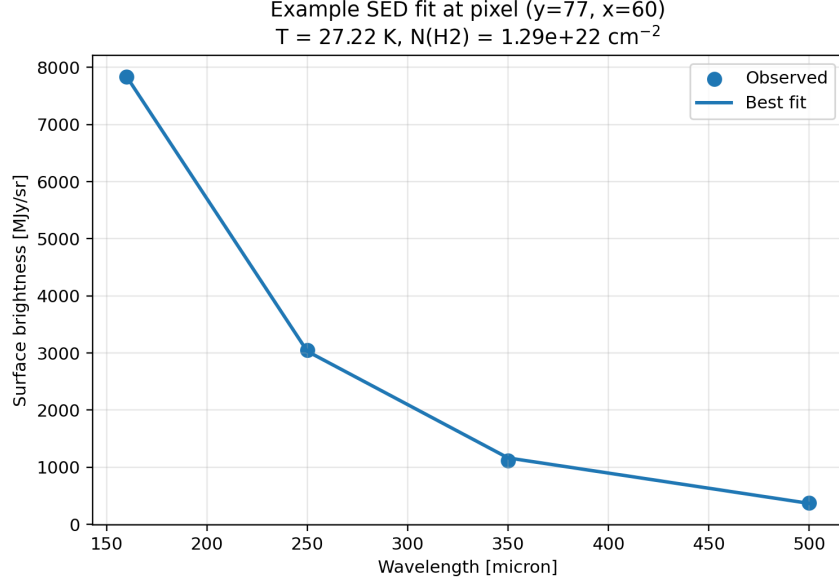


Figure 5: Example modified blackbody fit for one bright valid pixel. The four Herschel intensities are fitted using a fixed emissivity index $\beta = 2$.

6 Results

6.1 Fit statistics

The fitting was performed on all pixels satisfying the common mask. The fit statistics are summarized in Table 2.

Table 2: Summary of pixel by pixel SED fitting.

Quantity	Value
Total pixels in final grid	22320
Valid fitted pixels	4896
Failed fits	0
Fitted wavelength bands	160, 250, 350, 500 μm
Fixed emissivity index	$\beta = 2$
Temperature bounds	5 to 60 K
$\log_{10} N(\text{H}_2)$ bounds	18 to 25

6.2 Dust temperature map

The fitted dust temperature map is shown in Fig. 6. The fitted temperature values are summarized in Table 3.

Table 3: Dust temperature results.

Statistic	Temperature
Minimum	6.868 K
Maximum	60.000 K
Median	18.423 K

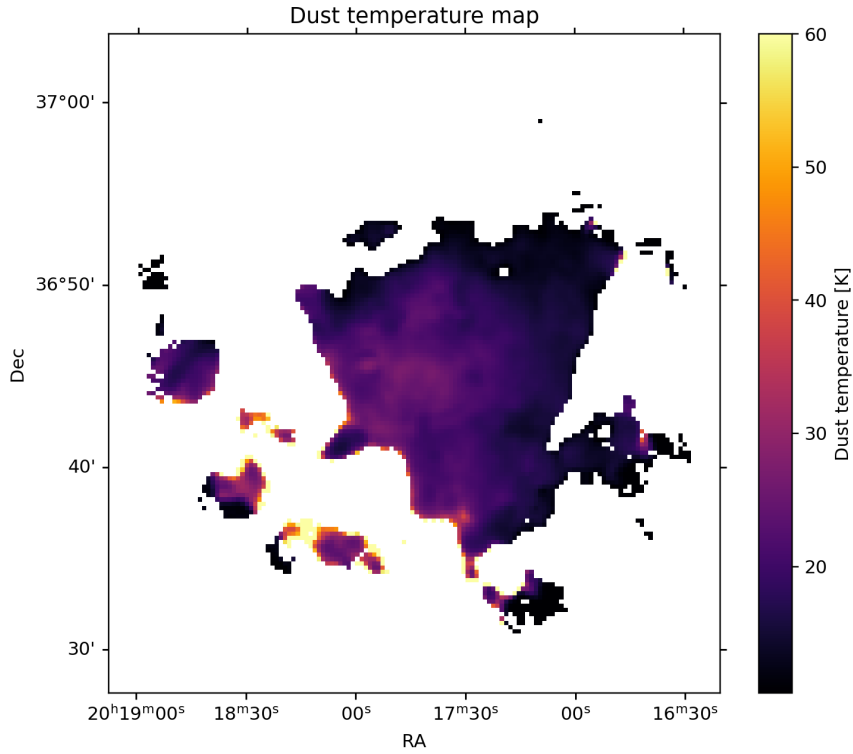


Figure 6: Dust temperature map obtained from pixel by pixel modified blackbody fitting. The warmest pixels reach the upper fitting bound of 60 K and should be interpreted with caution.

The temperature map traces the thermal state of the dust. Colder regions are usually associated with denser and more shielded parts of the molecular cloud. Warmer regions may indicate heating by embedded sources, nearby stars, or stronger external radiation fields. The physical interpretation should be made carefully because temperature can also be affected by line of sight mixing and map calibration.

6.3 Column density map

The fitted molecular hydrogen column density map is shown in Fig. 7. Since the column density spans several orders of magnitude, the map is plotted as $\log_{10} N(\text{H}_2)$.

Table 4: Molecular hydrogen column density results.

Statistic	$N(\text{H}_2)$
Minimum	$1.975 \times 10^{18} \text{ cm}^{-2}$
Maximum	$1.345 \times 10^{22} \text{ cm}^{-2}$
Median	$1.276 \times 10^{21} \text{ cm}^{-2}$

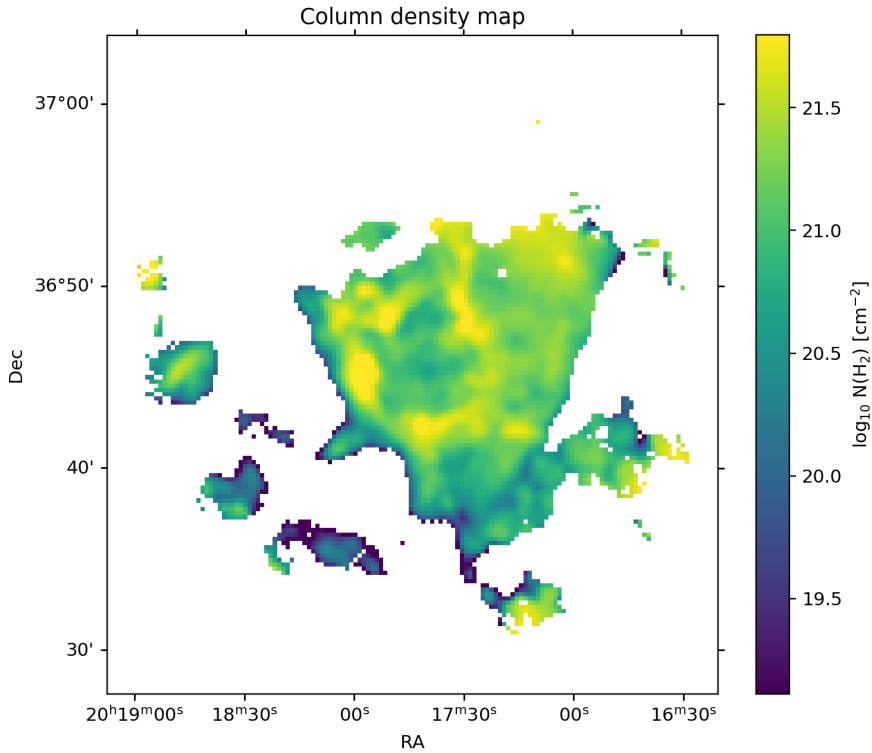


Figure 7: Logarithmic molecular hydrogen column density map derived from the modified blackbody fit. High column density regions trace the denser parts of the cloud.

The column density map traces the amount of molecular material along the line of sight. Regions with high $N(\text{H}_2)$ correspond to denser cloud structures and are possible locations of dense cores or filaments. Since the model assumes optically thin emission, fixed opacity, and fixed β , the absolute column density scale depends on these assumptions.

6.4 Relative residual map

The relative residual map is shown in Fig. 8. It gives a basic diagnostic of how well the modified blackbody model fits each pixel.

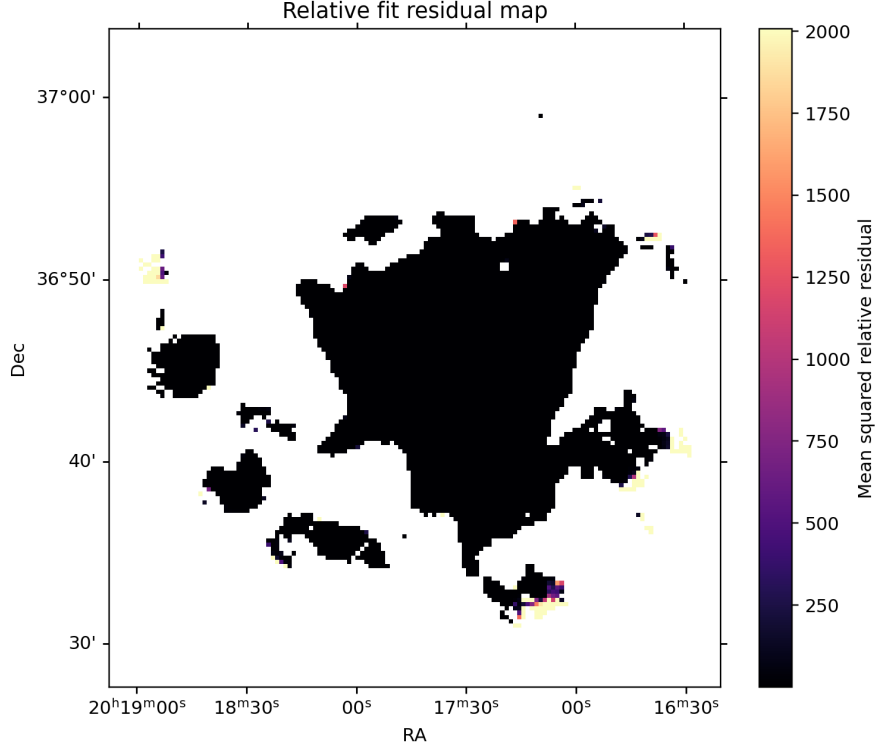


Figure 8: Relative residual map from the modified blackbody fitting. Larger values indicate pixels where the four band SED is less well represented by the simple fixed β model.

A high residual does not automatically mean the result is wrong. It can arise from noise, background subtraction effects, calibration uncertainties, temperature variations along the line of sight, or the assumption of a fixed emissivity index.

7 Physical Interpretation

The final maps show how multi wavelength far infrared and submillimetre observations can be converted into physical cloud properties. The temperature map gives the distribution of dust heating, while the column density map gives the projected distribution of material. In general, regions with high column density are physically important because they mark where the cloud is denser and more shielded. Such regions are more likely to be related to dense cores or filamentary cloud structure.

The method works because cold dust emission is approximately optically thin in this wavelength regime. Under this condition, the observed surface brightness is proportional to both the Planck function and the optical depth. The Planck function carries the temperature dependence, while the optical depth carries the information about the amount of dust and gas. By fitting several wavelengths at once, the degeneracy between temperature and column density is reduced.

The main physical limitations are as follows. First, the opacity κ_ν is assumed, not measured directly. Changing the opacity normalization changes the absolute column density. Second, the emissivity index β was fixed at 2. If the real dust grains have a different emissivity law, the fitted temperatures and column densities will shift. Third, each pixel may contain dust at multiple temperatures along the line of sight, while the model assumes a single effective temperature. Fourth, the maximum fitted temperature reaches the imposed upper bound of 60 K for some pixels, so those high temperature pixels should not be over interpreted.

8 Conclusion

In this project, Herschel maps at 160 μm , 250 μm , 350 μm , and 500 μm were used to estimate dust temperature and molecular hydrogen column density. The maps were first converted to common surface brightness units, convolved to the common 500 μm beam, reprojected to the same WCS grid, and masked to retain only pixels valid in all four bands. A modified blackbody model was then fitted to the four band SED at each valid pixel.

The final fit used 4896 valid pixels and had zero failed fits. The median dust temperature was 18.423 K, and the median molecular hydrogen column density was $1.276 \times 10^{21} \text{ cm}^{-2}$. The final temperature and column density maps provide a physically interpretable view of the thermal and mass structure of the cloud. The results are reliable as a first order dust based estimate, but the absolute column density scale depends on the assumed dust opacity and emissivity index.

Code Availability

The project was implemented using six Python scripts:

1. `00_inspect_fits.py`
2. `01_extract_convert_preview.py`
3. `02_convolve_to_500beam.py`
4. `03_reproject_to_500grid.py`
5. `04_make_common_mask.py`
6. `05_fit_modified_blackbody.py`

The final FITS products were saved in `data/processed/sed_fitting/`, and the report figures were saved in `plots/`.

References

- [1] Kauffmann, J., Bertoldi, F., Bourke, T. L., Evans II, N. J., and Lee, C. W. (2008). *MAMBO mapping of Spitzer c2d small clouds and cores*. *Astronomy & Astrophysics*, 487, 993 to 1017. <https://doi.org/10.1051/0004-6361:200809481>
- [2] Ossenkopf, V. and Henning, T. (1994). *Dust opacities for protostellar cores*. *Astronomy & Astrophysics*, 291, 943 to 959.
- [3] Pilbratt, G. L. et al. (2010). *Herschel Space Observatory: An ESA facility for far infrared and submillimetre astronomy*. *Astronomy & Astrophysics*, 518, L1. <https://doi.org/10.1051/0004-6361/201014759>
- [4] Aniano, G., Draine, B. T., Gordon, K. D., and Sandstrom, K. (2011). *Common resolution convolution kernels for space and ground based telescopes*. *Publications of the Astronomical Society of the Pacific*, 123, 1218 to 1236. <https://doi.org/10.1086/662219>